

Anders Nilsson,
Ingenjörsvetenskap
0702-23 83 79
ander.nilsson@hv.se

Washing machine, controlled by a combined logic and SFC

Purpose

- Develop your logical thinking in FBD
- Try sequential programming in SFC
- Practise combining FBD and SFC and the use of internal communication mx variables

Preparation

- Download TwinCAT Project Washer from Canvas

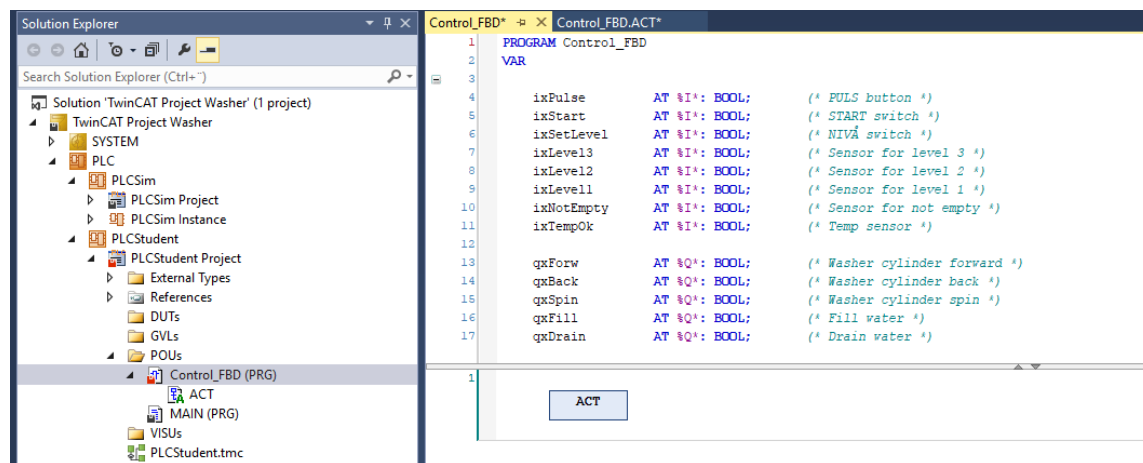
Exercise 1

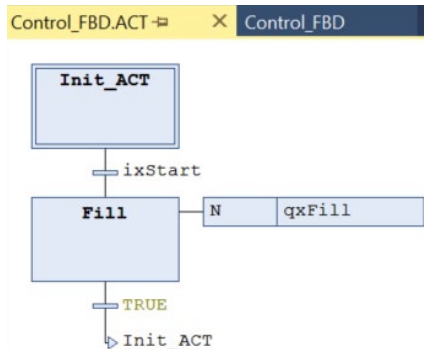
Create a Sequential Function Chart (SFC) that fills the washing machine when the start knob is on. The machine should be filled to level 2. After that, turn 2 seconds forward and then two seconds back. Then empty the machine, and then the centrifuge should run for 5 seconds.

- There is a time measure function built in SFC for use in a transition, e.g., `actionName.t > T#2s`. Make a transition to the next SFC step after 2 seconds.
- The centrifuge function is engaged if the two output variables `qxForw` and `qxSpin` are TRUE at the same time.

Instructions on how to add an SFC Action to your existing Control_FBD(PRG):

- Right-click on Control_FBD and select "Add" and "Action"
- Change the language to "Sequal Function Chart"
- Click on Open, and you should see an empty SFC
- Build your SFC program by right-clicking on a transition and selecting "insert step-transition after"
- Right-click on the Step squares and select "Insert Action Association" to assign variables
- Call your new SFC action from FBD by
 - Right-click in a blank "Network" in the FBD and select "Insert empty Box"
 - Write "ACT" if that is the name of the SFC action you created.



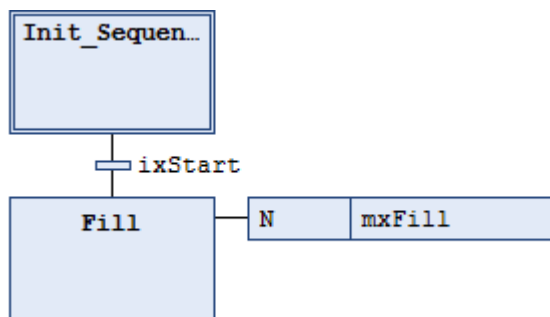


Exercise 2

Create a Paus function for the washing machine using SFC and FBD.

Change the output variables in your SFC to internal Boolean variables.

Ex. Change qxFill to mxFill. A dialog box should appear that asks you to create a new variable, check that the variable type of BOOL is selected, and press OK.



Add AND functions into your Control_FBD that assign the output variables to TRUE if ixStart is TRUE, and in this case, mxFill is TRUE.

Change the conditions for all included output variables in the same manner.

Test run the solution, turn the start knob off, and check that the machine pauses. Then, turn it on, and the machine will continue to operate as usual. Do not care that the timers will run out during the pause.

Take screenshots of your SFC and FBD solutions and submit them to Canvas.